The Problem Statement:

The Alpha-Wave Brain Computer Interface Front-End

The human brain emits distinct electrical patterns based on mental states. Alpha waves (**8 Hz to 12 Hz**) are prominent when a person is relaxed with their eyes closed. However, these signals at the scalp are incredibly weak (**10 μV to 50 μV**) and are easily drowned out by the **50 Hz** powerline interference radiating from every wall outlet, as well as baseline wander from patient movement.

The Task: Design a single-channel, wearable Analog Front-End (AFE) that isolates and amplifies Alpha brainwaves from raw scalp electrode inputs. The output must be a clean, amplified signal ready to be fed into an standard 3.3V ADC (like an ESP32 or Arduino) for relaxation detection.

Design Specifications:

- **Total System Gain:** 1000 to 5000 V/V (distributed appropriately to prevent rail saturation).
- **Frequency Response (Bandpass):** Strictly pass the **8 Hz to 12 Hz** Alpha band. Frequencies below **0.5 Hz** (DC offset/sweat artifacts) and above **30 Hz** (muscle noise/EMG) must be heavily attenuated.
- **Noise Rejection:** Must include a dedicated stage to aggressively reject **50 Hz** mains hum (minimum -40 dB attenuation at 50 Hz).
- **Input Stage:** Must handle differential inputs from three electrodes (Active, Reference, and Right Leg Drive/Ground) with a very high Common-Mode Rejection Ratio (CMRR).
- **Power Supply:** The circuit must operate on a single-supply (e.g., 3.3V or 5V) or dual-supply (+/- 5V or +/- 9V) derived from batteries, fitting a wearable form factor.
- **PCB Constraints:** Maximum board size of **100mm x 100mm**.

Design Guidelines & Hints:

1. **Input Stage:** Consider how your circuit will extract a 10μV differential brainwave when the human body acts as a giant antenna picking up massive 50 Hz common-mode noise.
2. **Gain Distribution:** The skin-electrode interface naturally generates its own DC voltage; applying all your system gain in a single stage will instantly saturate your op-amps to the power rails.
3. **Noise Management:** Instead of just filtering 50 Hz mains hum after it enters the circuit, research how professional biopotential systems actively cancel noise on the user's body first.
4. **Filter Precision:** A basic RC filter won't suffice; you need a sharp enough topology to isolate the tight 8–12 Hz Alpha band without bleeding in motion artifacts or muscle noise.
5. **Output Biasing:** Brainwaves naturally swing positive and negative, but standard microcontrollers cannot read negative voltages; plan your final stage so the signal is centered for the ADC.

Deliverables: **P1:** Simulation Instructions are attached, the Phase 1 Submission should involve output of the test mentioned , along with a report on the designed circuit justifying component choice.

P2: The Spice models will be tested with instantaneous testbenches on the location , the detailed presentation with simulation results and PCB layout to be presented in front of the Judges.